Proof of Euler's Identity

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1 Proof

First let's recall Euler's identity: $e^{i\pi} = -1$

To prove this identity, we will need to use important Maclurin series definitions of the functions e^x, $\sin x$, and $\cos x$.

Maclurin Definitions:

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \cdots +$
- $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots +$
- $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \cdots +$

The first step to proceed with this proof, is to write out the Maclurin expansion of e^x:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \cdots +$$

For the next step, we will write out the Maclurin expansion of e^{ix}:

$$e^{ix} = 1 + ix + \frac{(i)^2x^2}{2} + \frac{(i)^3x^3}{6} + \frac{(i)^4x^4}{24} + \frac{(i)^5x^5}{120} + \cdots +$$

We know that $i = \sqrt{-1}$, and $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and so on. Using this information, let's expand the terms in the Maclurin series above.

$$e^{ix} = 1 + ix + \frac{-1x^2}{2} + \frac{-ix^3}{6} + \frac{x^4}{24} + \frac{ix^5}{120} + \cdots +$$

$$\implies e^{ix} = 1 + ix - \frac{x^2}{2} - \frac{ix^3}{6} + \frac{x^4}{24} + \frac{ix^5}{120} + \cdots +$$

Our next step is to group all the terms with i together, and group those not including i.

$$e^{ix} = \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots\right) + i\left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots\right)$$

Notice that the terms without i correspond to the Maclurin series for $\cos x$, and the terms with i correspond to the Maclurin series for $\sin x$. Therefore, we have:

$$e^{ix} = \cos x + i \sin x$$

This is Euler's formula. Now, to prove Euler's identity, substitute $x = \pi$:

$$e^{i\pi} = \cos \pi + i \sin \pi$$

We know that $\cos \pi = -1$ and $\sin \pi = 0$. Thus:

$$e^{i\pi} = -1 + i(0)$$

$$\implies e^{i\pi} = -1$$

This completes the proof of Euler's identity.
